



Visual Analytics System for Data Quality Improvement

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Introduction

- In this paper, we propose a novel visual analytics system to manage **data quality** to improve the performance of **machine learning models**.
- The proposed system suggests an **optimal quality improvement method and process** for data to be appropriately used in models.

Challenges & Design goal

C1

Existing visual analytics systems are only focused on improving data quality.

C2

Different techniques and process are used to improve data quality depending on data and purpose.

D1

System needs to support evaluation metrics and data quality improvement processes for various machine learning models.

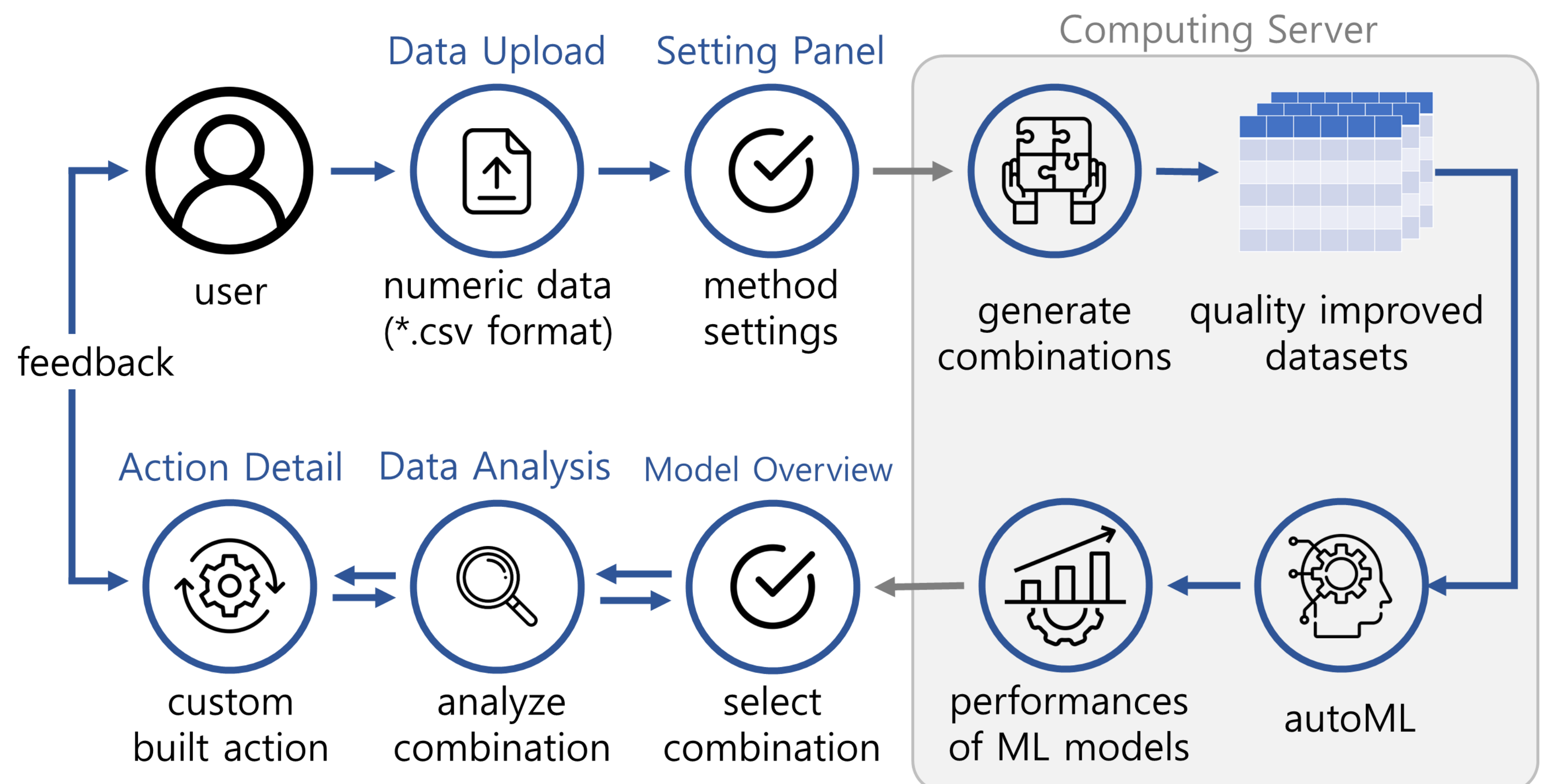
D2

System needs to provide the basis for the optimal data quality improvement process.

D3

System needs to provide functions for users to improve data quality in detail.

System architecture



System & Case study

- We aim to improve data quality for machine learning models that predict prices.
- We generate 10% of missing and inconsistent values in dataset for the case study and do not generate outliers because they exist sufficiently.

setting panel

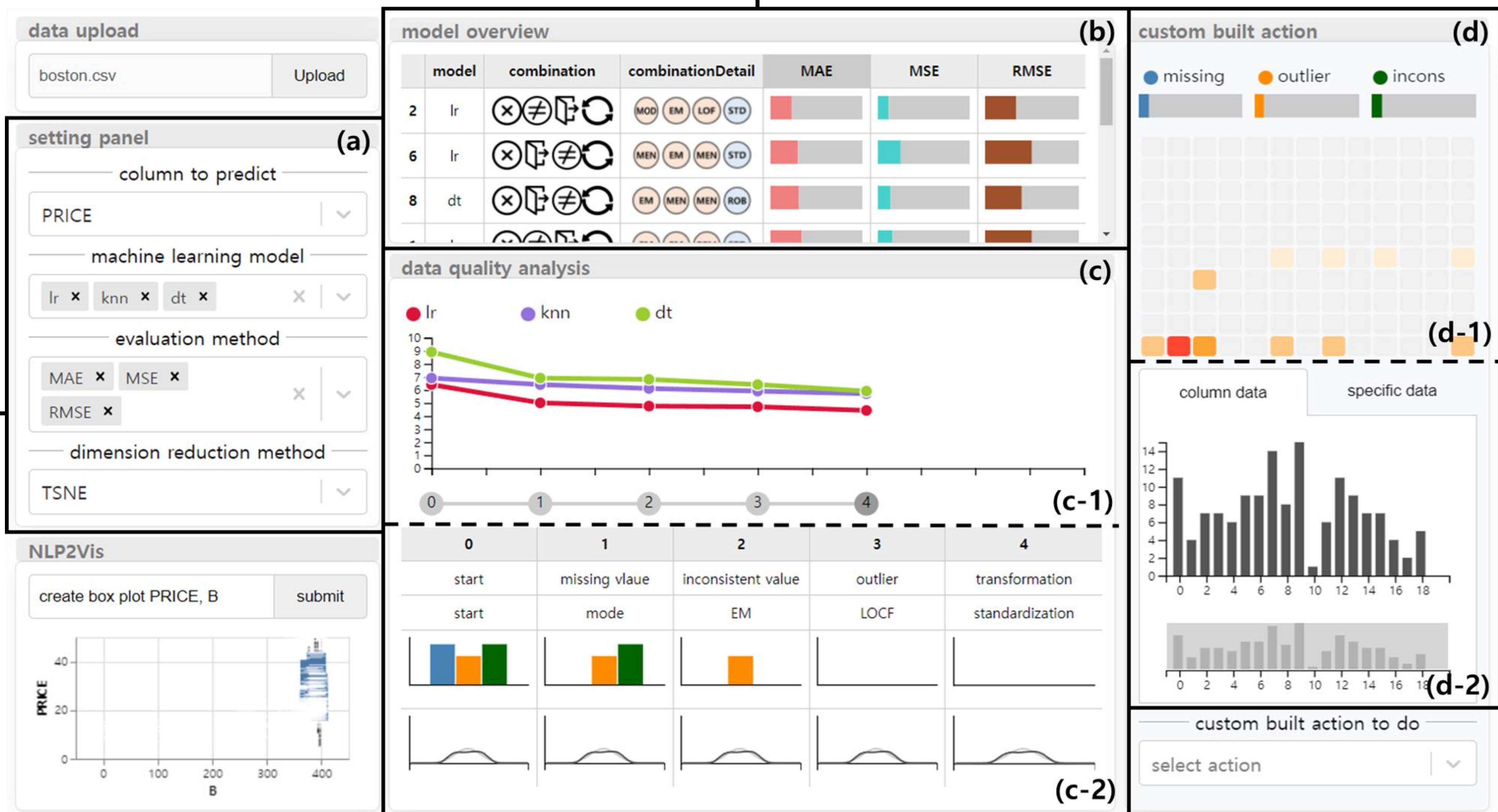
We **select models and performance evaluation metrics** which mainly used to predict time series data.

model overview

System show the model **performance and process for each combination** using icons.

Custom action built

We perform the **detailed customization** to not apply the data quality improvement method to the outlier of the price column.

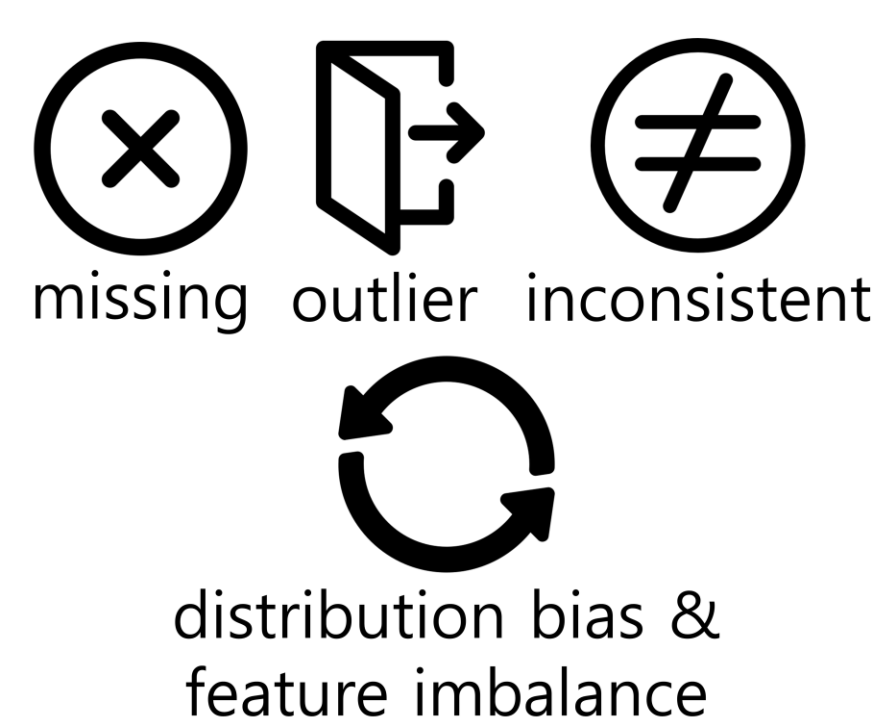


Data quality analysis

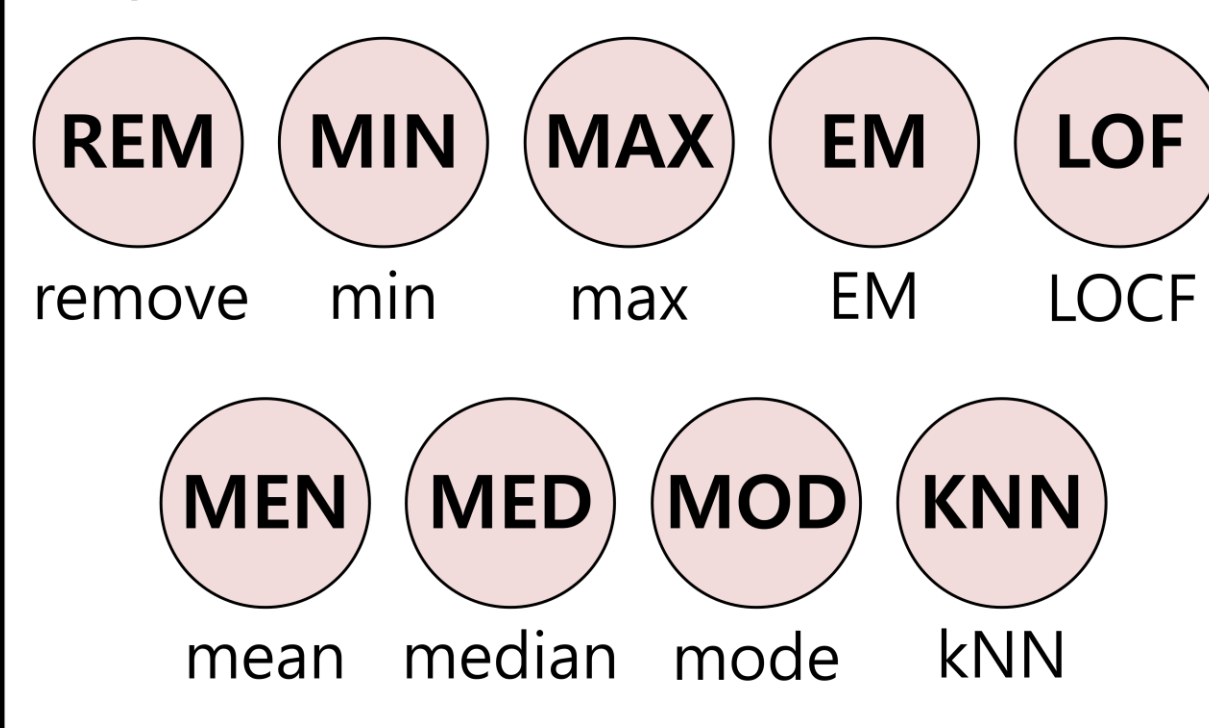
System shows the **evidence** that the combination is the optimal data quality improvement process.

Icons used by the system

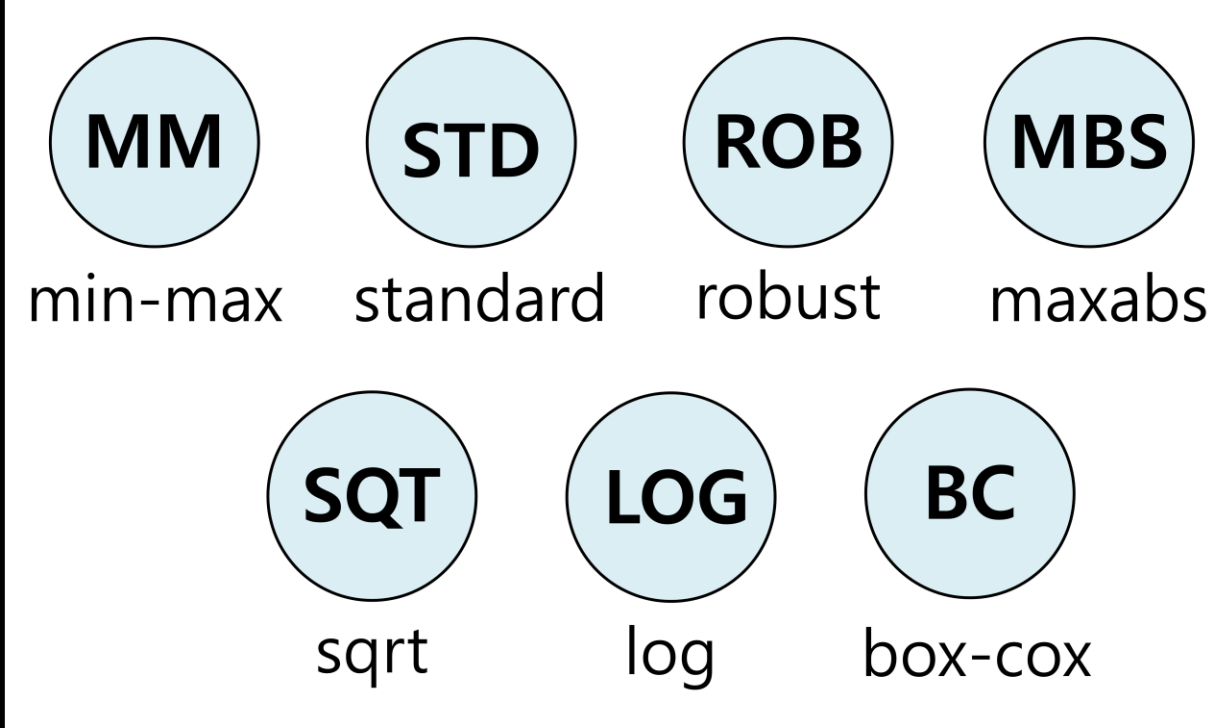
cause of poor quality



imputation methods



transformation methods



Conclusion & Future work

- We introduced system, which supports decision-making while improving data quality.
- We plan to investigate more techniques besides imputation and transformation to enhance data quality in future work.
- We will study data quality improvement techniques for classification problems.